Numbers That Talk

Divisibility shortcuts, and the modular arithmetic hiding behind them

Zhiheng Li · Yueran · MATH 5248 · December 2026

Do this with us during the talk. The **why** is on the back.

Warm-up. This year is 2026. Is it divisible by 9? Add the digits: 2 + 0 + 2 + 6 = _____. Is that a multiple of 9? ☐ yes ☐ no. So the remainder of 2026 ÷ 9 is _____.

1 The vanishing digit (Zhiheng)

1. Think of a four-digit number. Not all four digits the same. Don't tell us. **My number:** _____
2. Shuffle its digits into a new number. **Shuffled:** _____
3. Subtract the smaller one from the bigger one. _____ − _____ = _____
4. Circle **one non-zero digit** of the result. Keep it hidden.
5. Read us the other digits, in any order. We name the hidden one. **We said** ____ ☐ right

Example. 5274 → shuffle 7425 → 7425 − 5274 = 2151 → hide the 5 → you read 2, 1, 1 → we say **5**.

2 1001 brings it back (Yueran)

1. Think of a three-digit number. **My number:** _____
2. Write it twice, so it becomes a six-digit number. _____
3. Divide by 7. = _____ (it divides evenly, promise)
4. Divide that by 11. = _____
5. Divide that by 13. = _____ ☐ my number is back

Example. 572 → 572572 → ÷ 7 = 81796 → ÷ 11 = 7436 → ÷ 13 = **572**.

3 Try it after the talk

1. Is 45658 divisible by 37? Cut it into three-digit chunks from the right: 45 | 658. Add: 45 + 658 = _____. Is that a multiple of 37? ☐ yes ☐ no
2. What is 2026 mod 7? Chunks 2 | 026. Alternate: 026 − 2 = _____. That mod 7 is ____, and mod 13 it is ____.
3. 1/7 = 0.142857 142857… Multiply 142857 by 2: _____. By 3: _____. Same digits?

Full proofs, 142857, Midy, a quick rule for 7 and ISBN: **github.com/ZhihengLi0/math5248**

Why does any of this work? → turn over

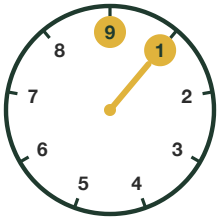
Why it works

Everything below uses only tools from MATH 5248: congruences, Fermat and Euler, the order of an element. Full proofs, every example and a checker script: **github.com/ZhihengLi0/math5248**

A · Trick 1 is "10 ≡ 1 (mod 9)"

Theorem. For any n with digits d_0, d_1, … (units first), $n \equiv d_0 + d_1 + \dots \pmod 9$, hence also mod 3.

Proof. $n = d_0 + d_1 \cdot 10 + d_2 \cdot 10^2 + \dots$. Since $10 = 9 + 1$, $10 \equiv 1 \pmod 9$, so $10^i \equiv 1$ and $d_i \cdot 10^i \equiv d_i$. Add up. 3 | 9 gives the mod 3 version. ∎



A clock with only 9 marks. 10 steps land where 1 step lands. So do 100 and 1000.

The trick. Your number n and the shuffled n′ have the same digits, so $n \equiv n' \pmod 9$ and 9 | n − n′. By the theorem the digits of the difference add to a multiple of 9. You read all but one digit x; the sum s you read satisfies $s + x \equiv 0 \pmod 9$, and exactly one x in 1…9 does that. (A 0 is banned because 0 and 9 would look the same.)

2026 → 2+0+2+6 = 10 → 1, so $2026 \equiv 1 \pmod 9$. 7452 → 18, so 9 | 7452.

B · Trick 2 is "1000 ≡ −1 (mod 1001)"

abcabc = abc·1000 + abc = abc·1001, and **1001 = 7 × 11 × 13**. Dividing by 7, 11, 13 in turn is dividing by 1001, so abc comes back. It cannot fail.

The real point: $1000 = 1001 - 1$, so $1000 \equiv -1 \pmod{1001}$. Cut n into three-digit blocks B_0, B_1, B_2, … from the right. Then $n = B_0 + B_1 \cdot 1000 + B_2 \cdot 1000^2 + \dots \equiv B_0 - B_1 + B_2 - \dots \pmod{1001}$, hence mod 7, 11 and 13 at once.

2026 → 026 − 2 = 24 → $24 \equiv 3 \pmod 7$, $\equiv 11 \pmod{13}$, $\equiv 2 \pmod{11}$. All three are the true remainders.

C · One idea, every rule

Block theorem. If $10^k \equiv 1 \pmod d$, then $n \equiv B_0 + B_1 + B_2 + \dots \pmod d$ for k-digit blocks from the right. If $10^k \equiv -1$, then $n \equiv B_0 - B_1 + B_2 - \dots$. Proof: write n in base 10^k and use $(10^k)^j \equiv (\pm 1)^j$, exactly as in A. ∎

d	k	$10^k \equiv$	rule	example
3, 9	1	+1	digit sum	2026 → 10 → 1
11	1	−1	alternate digits	2026 → 6−2+0−2 = 2
101	2	−1	alternate 2-blocks	461267 → 67−12+46 = 101 ✓
27, 37	3	+1	add 3-blocks (999 = 27×37)	45658 → 45+658 = 703 = 19×37 ✓
7, 13, 77	3	−1	alternate 3-blocks	1283945 → 945−283+1 = 663 = 13×51 ✓
41, 271	5	+1	add 5-blocks (99999 = 9×41×271)	$\text{ord}_{41}(10) = 5$

D · Every prime p ≠ 2, 5 gets a rule

Theorem. (1) (p − 1)-digit blocks, added, always work. (2) The shortest *additive* block is $k = \text{ord}_p(10)$, and k | p − 1. (3) If k is even, $10^{k/2} \equiv -1 \pmod p$, so (k/2)-blocks with alternating signs work too.

Proof. (1) p does not divide 10, so Fermat gives $10^{p-1} \equiv 1 \pmod p$; apply the block theorem. (2) By definition of the order, k is the least exponent with $10^k \equiv 1$, and $a^n \equiv 1 \Rightarrow \text{ord} \mid n$ gives k | p − 1. (3) Put $x = 10^{k/2}$. Then $p \mid x^2 - 1 = (x - 1)(x + 1)$. By Euclid's lemma $p \mid x - 1$ or $p \mid x + 1$; the first contradicts the minimality of k. ∎

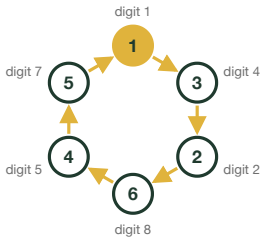
Careful. (3) needs p prime: for d = 33, ord(10) = 2 but 10 is not ≡ −1 (mod 33). For composite d with gcd(d, 10) = 1, Euler gives $10^{\phi(d)} \equiv 1$, so φ(d)-blocks work and the shortest additive block is still $\text{ord}_d(10)$. Example: φ(27) = 18 but $\text{ord}_{27}(10) = 3$, since 999 = 27 × 37.

7: ord = 6 | 6, even, so $10^3 \equiv -1$. 13: ord = 6 | 12. 37: ord = 3 | 36 (odd, no −1 rule). 41: ord = 5 | 40.

E · Block length = period of 1/p

Theorem. If gcd(d, 10) = 1, d > 1, then 1/d is purely repeating with period $\text{ord}_d(10)$.

Proof. Long division starts with remainder $r_0 = 1$ and each step replaces r by 10r mod d, so $r_i \equiv 10^i \pmod d$. The digits repeat from the first i with $r_i = 1$, which is $i = \text{ord}_d(10)$. ∎ Also directly: $10^6/7 - 1/7 = 142857$, so $999999 = 7 \times 142857$.



Remainders of 1 ÷ 7: 1, 3, 2, 6, 4, 5, back to 1. They are 10^i mod 7. Six steps, so six digits.

So "how many digits per block for 7" and "how long is the period of 1/7" are the same question. Both answers are 6.

Bonus. 142857 × 1, 2, …, 6 = 142857, 285714, 428571, 571428, 714285, 857142: the same digits rotated, and × 7 = 999999. Reason: the powers of 10 mod 7 hit every non-zero remainder, so 10 is a *primitive root* mod 7, and k/7 just enters the same cycle at a different point. **Midy:** 142 + 857 = 999 and 076 + 923 = 999 (for 1/13), because $10^3 \equiv -1$.

F · Not in the talk, in the full deck

Chop-the-last-digit rules from 10^{-1} mod d (7: x − 2y; 13: x + 4y). 2^k and 5^k: last k digits, since $10 \equiv 0$. Coprime factors combine (6 = 2 × 3), non-coprime do not (4 and 6 divide 12, but 24 does not). Other bases: digit sum tests b − 1, alternating sum tests b + 1. ISBN-10 check digits use mod 11 because 11 is prime.

References

MATH 5248 notes: congruences (Day 3), units and inverses (Day 4–5), Euclid's lemma (Day 6), CRT (Day 10–11), Fermat (Day 11–12), Euler (Day 12–13), order (Day 13), primitive roots (Day 14). Hoffstein, Pipher, Silverman, *An Introduction to Mathematical Cryptography*, pp. 30–32. NTIC (Number Theory: In Context and Interactive). Wikipedia: Divisibility rule; Repeating decimal; Multiplicative order; Cyclic number; Midy's theorem. Every number here was checked by our script *verify/check_examples.py* in the repo.